

Carolina Del Pilar Díaz

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Summary

AI/ML Engineer, Senior Data Scientist and Applied Statistician (Economist) with 20+ years of experience, operating as a Forward Deployed Engineer (FDE) and Technical Lead for enterprise clients. I architect and ship end-to-end GenAI products — Multi-Agent Systems, RAG and MCP pipelines — with the rigor of Advanced Statistics, Causal Inference and MLOps, on on-prem, hybrid and cloud (AWS, Azure, Databricks) infrastructures across Fintech, AgTech and Healthcare.

Experience

IT Patagonia (Consultant FDE), Senior AI Engineer / Forward Deployed Engineer

Buenos Aires, Argentina

June 2024 – present

- FDE & Technical Lead: architected and deployed GenAI, RAG and Multi-Agent solutions for enterprise clients (Banco Galicia, ICBC, Banco Macro, Facturante), defining technical roadmaps and cloud architecture.
- Agentic AI (Patagon.IA): 33-node LangGraph state-machine on AWS Bedrock for automated COBOL-legacy BRD/TRD documentation — led prompt engineering (~23 sections) and a 3-judge LLM majority-voting panel with anti-hallucination filters vs. a deterministic golden-truth (20/20 sections verified, 0 pipeline errors).
- Production GenAI apps: WhatsApp sales agent as an MLflow PyFunc model on Databricks (RAG via Vector Search + Gemini); two-agent invoice-extraction & eligibility platform on AWS Lambda + Bedrock (Claude Sonnet/Haiku, Strands framework); React 19 + Gemini conversational data apps.
- LLM cost & quality engineering: state-checkpointing to avoid API reprocessing, strict token budgets, dynamic cloud/local routing (AWS Bedrock / Ollama), and an LLM-as-a-judge multi-evaluator with majority voting for hallucination control.

ITP Patagonia - ICBC, Senior Data Scientist

Buenos Aires, Argentina

June 2024 – July 2025

- Built an RFM customer segmentation on AWS SageMaker Canvas / Data Wrangler over Amazon Athena (customer-360 join, ~250 features, ~1M monthly records) with robust outlier handling, similarity (MinHash) encoding and STL/LOESS decomposition.
- Developed segmentation and investment-propensity models for mutual funds (FCI) and a credit-card churn classifier (Random Forest, AdaBoost, SVM; SMOTE, cross-validation, F1).
- Designed a hybrid fraud-detection approach combining a rules engine with anomaly detection (Isolation Forest, Autoencoders, One-Class SVM).

Nubiral S.A., Senior Data Scientist — COE, Generative Models

Buenos Aires, Argentina

Oct 2023 – May 2024

- Integrated AWS Rekognition, Lambda, S3, LangChain and OpenAI GPT for Computer Vision + Generative AI at Aerolíneas Argentinas; built a Post-Call Analytics solution (Transcribe, Comprehend, QuickSight).
- Implemented Azure Cognitive Search and Form Recognizer for PDF document management at Banco Macro.

Pi Data Strategy & Consulting, Semi-Senior Data Scientist / Statistician / Economist

Córdoba, Argentina

Mar 2021 – Oct 2023

- End-to-end predictive and regression modelling (EDA, classification, neural networks) deployed on Azure; sales forecasting and customer-satisfaction models for AB InBev, Don Mario and Pernod Ricard.

National Institute of Agriculture Technology (INTA), Lead Statistician / Head of Statistical Research Manfredi, Córdoba, Argentina
Jan 2012 – Dec 2021

- Designed experimental designs and mixed statistical models across research programs; applied neural networks, deep learning and computer vision to UAV and agricultural-machinery crop monitoring.

Skills

Agentic & Generative AI: Multi-Agent Systems, LangGraph, LangChain, LlamaIndex, Strands, AutoGen, Semantic Kernel, RAG, MCP, Prompt Engineering, LLMs (Gemini, GPT, Claude, DeepSeek, Ollama, Hugging Face)

Machine Learning & Stats: Deep Learning, Computer Vision, Causal Inference, Bayesian Statistics, Econometrics, Anomaly Detection, Time Series, MLOps

Cloud & Big Data: AWS (SageMaker, Bedrock, Lambda, Rekognition), Azure AI Services, Databricks, Snowflake, Apache Spark, Docker, FastAPI, Python, R, SAS

Languages: Spanish (Native), English (Intermediate), Portuguese (Basic)

Projects

PATAGON.IA — Automated COBOL Legacy Documentation Jan 2026

33-node LangGraph pipeline on AWS Bedrock generating BRD/TRD docs from COBOL source; ~23-section prompt engineering, a 3-judge LLM panel (majority voting) and anti-hallucination filters vs. a deterministic golden-truth. Human-in-the-loop and multi-model (DeepSeek R1 / Qwen / GLM). IT Patagonia.

Banco Macro — Nightly Batch Completion Predictor (OPTI+) 2025-2026

Statistical predictor of batch end-time as an hour window. Compared Linear Regression, Random Forest, HistGradientBoosting and Prophet (MAE/R² over 485 records) plus an instance-based k-NN estimator with robust median and asymmetric ranges; leave-one-out over 105 days, 99% within tolerance+SLA. FastAPI + React + Docker. IT Patagonia.

Facturante — WhatsApp Sales AI Agent Jan 2026

Conversational sales agent deployed as an MLflow PyFunc model on Databricks Model Serving with RAG (Vector Search) and Gemini. Automated LLM-as-judge evaluation over a golden dataset, consistency scoring and Lakeview conversation auditing. IT Patagonia.

ICBC — Banking Customer Segmentation (RFM) Jan 2025

RFM segmentation on AWS SageMaker Canvas/Data Wrangler over Amazon Athena (~250 features), with robust- σ outlier handling, similarity (MinHash) encoding and STL/LOESS decomposition over ~1M monthly records. IT Patagonia.

Banco Galicia — AI Architecture & Observability Assessment 2025-2026

Code-based audit of two production AI systems (Java/Kafka/OpenShift multi-agent orchestrator + Azure OpenAI RAG). Designed the observability strategy (Prometheus, Grafana, OpenTelemetry/Jaeger, RAGAS) and a target architecture aligned to BCRA regulation. IT Patagonia.

ITP Flokzu — Agentic Document Platform Jan 2026

Two-agent platform on AWS Lambda (Python 3.13) + Bedrock (Claude Sonnet/Haiku, Strands) for invoice field extraction and Knowledge-Economy eligibility decisions, integrated with Flokzu BPM; CloudWatch monitoring. IT Patagonia.

ICBC — Legacy Modernization VB6 → C#/.NET 8 Jan 2026

Migration of the judicial-orders banking system (AFIP/BCRA integration) from VB6 to a layered .NET 8 architecture with EF Core, Dependency Injection, async/await, Serilog and DB2. IT Patagonia.

Education

Postgraduate Degree	Master's National University of Córdoba, Applied Statistics	Córdoba, Argentina Jan 2006 – Dec 2008
Diploma	National University of Córdoba, Data Science, Machine Learning and applications	Córdoba, Argentina Jan 2019 – Dec 2020
Bachelor's degree	National University of Córdoba, Economics	Córdoba, Argentina Jan 1993 – Dec 1998

Certifications

- Databricks Certified Generative AI Engineer Associate (2026)
- Microsoft Certified: Azure AI Fundamentals — Candidate (2026)
- Agentic AI on AWS — BeSA (Become a Solution Architect), Candidate (2026)
- AWS Academy Graduate — Cloud Foundations & Machine Learning Foundations (2024–2026)
- Bayesian Statistics: Techniques and Models — University of California (2021)

Publications

- 20+ peer-reviewed publications in applied statistics, genomics and agronomy (2014–2020). Full list available on request.